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1. Three planes have equations

$$x - y + 2z = 3$$

$$2x + y + z = 4$$

$$3x + (m + 1)z = 7$$

where m is a constant.

(i) Investigate the arrangement of the planes:

when $m = 2$;

when $m \neq 2$.

[6]

(ii) A student claims that the position vectors $\mathbf{i} - \mathbf{j} + 2\mathbf{k}$, $2\mathbf{i} + \mathbf{j} + \mathbf{k}$ and $3\mathbf{i} + 4\mathbf{k}$ all lie in a plane through the origin. Determine whether or not the student is correct.

[2]

2. (a) Consider the linear system $A\mathbf{x} = \mathbf{b}$, where

$$A = \begin{pmatrix} 3 & -2 & 1 \\ 2 & 1 & -4 \\ 7 & 0 & -7 \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 5 \\ 1 \\ 7 \end{pmatrix}$$

Show that A is singular, and deduce that the system does not have a unique solution. [2]

- (b) Show that the system is consistent. Interpret the solution set geometrically. [3]

3. (a) Show that the system of equations

$$x + 2y - z = 3$$

$$2x - y + 4z = 1$$

$$3x + y + 3z = a$$

where a is a constant, does not have a unique solution.

[2]

- (b) Given that $a = 4$, show that the system of equations in part (a) is consistent. Interpret this situation geometrically.

[3]

- (c) Given instead that $a \neq 4$, show that the system of equations in part (a) is inconsistent. Interpret this situation geometrically.

[2]

4. The equations

$$x - y + 2z = 4$$

$$(k + 1)x + ky + (2 - k)z = 4 - k$$

$$(5 - 2k)x + (2 - 2k)y + (k + 2)z = 8 - k$$

represent three planes, where k is a real constant.

The planes do not intersect in exactly one point.

(a) Find the possible values of k . [3]

(b) For each value of k found in part (a), describe the configuration of the planes.
Fully justify your answer, stating in each case whether the corresponding system is consistent or inconsistent. [5]

5. Three planes have equations

$$\begin{aligned}x + ay &= -1 \\2x + y + z &= 3 \\3x + by + 2z &= c\end{aligned}$$

where a , b and c are constants.

(a) In the case where the planes **do not** intersect at a unique point,

(i) find b in terms of a

[4]

(ii) find the value of c for which the planes form a sheaf.

[3]

(b) In the case where $b = 1 - a$ and $c = 5$, find the coordinates of the point of intersection of the planes in terms of a .

[6]

6.

$$\mathbf{M} = \begin{pmatrix} 1 & 1 & 0 \\ 3 & k & 1 \\ 2 & -1 & 1 \end{pmatrix}$$

where k is a constant.

(a) Find the values of k for which the matrix $\mathbf{M}$ has an inverse. [2]

(b) Find, in terms of p , the coordinates of the point where the following planes intersect:

$$\begin{aligned} x + y &= 1 \\ 3x + 2y + z &= p \\ 2x - y + z &= 4 \end{aligned} \quad [5]$$

(c) (i) Find the value of q for which the set of simultaneous equations

$$\begin{aligned} x + y &= 1 \\ 3x + z &= q \\ 2x - y + z &= 4 \end{aligned} \quad [2]$$

has at least one solution.

(ii) For this value of q , interpret the solution set geometrically. [2]

7. You are given that a is a parameter which can take only real values.
The matrix A is given by

$$A = \begin{pmatrix} 3 & 2 & a \\ 1 & 3 & 0 \\ 2 & -1 & 1 \end{pmatrix}$$

- (a) Find an expression for the determinant of A in terms of a . [2]

You are given the following system of equations in x , y and z .

$$3x + 2y + az = 8$$

$$x + 3y = 5$$

$$2x - y + z = 3$$

The system can be written in the form

$$A \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 8 \\ 5 \\ 3 \end{pmatrix}$$

- (b) (i) In the case where A is not singular, solve the given system by using A^{-1} . [5]
(ii) In the case where A is singular, describe the configuration of the planes whose equations are the three equations of the system. [3]

The transformation represented by A is denoted by T .

A 3-D solid of volume $a^2 + 3$ is transformed by T to an image.

- (c) (i) Determine the range of values of a for which the orientation of the image is the reverse of the orientation of the object. [1]
(ii) Determine the range of values of a for which the volume of the image is less than the volume of the object. [2]